

(COMPUTATIONAL) INDISTINGUISHABILITY

ADVANCED TOPICS IN ~~CYBERSECURITY~~ CRYPTOGRAPHY (7CCSMATC)

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OUTLINE

Introduction

AES-CTR Mode

INTRODUCTION

ASSUMPTIONS

- We reached the limits of what can be done information-theoretically.
- Most cryptographic constructions give you conditional security: its security properties hold if and only if some computational task is hard.
- **Examples**
 - inverting AES is hard,
 - factoring large integers is hard, or
 - finding short solutions to a system of linear equations modulo q is hard.

What a proof gives you

The only way for an adversary **with these powers** to break **this specific security goal** is to solve this hard computational problem efficiently.

AES-CTR MODE

RECAP: IND-CPA AND AES-128-CTR

IND-CPA	$E(m_0, m_1 \in \{0, 1\}^{128 \cdot n})$	$E_k(m \in \{0, 1\}^{128 \cdot n})$
1: $k \leftarrow \$ \{0, 1\}^{128}$	1: if $ m_0 \neq m_1 $ then	1: $iv \leftarrow \$ \{0, 1\}^{128}$
2: $b \leftarrow \$ \{0, 1\}$	2: return $\perp$	2: $m_0, \dots, m_{n-1} \leftarrow m$
3: $b' \leftarrow \mathcal{D}^E$	3: $c \leftarrow \$ E_k(m_b)$	3: for $0 \leq i < n$ do
4: return $b = b'$	4: return c	4: $c_i \leftarrow \text{AES-128}(k, iv + i) \oplus m_i$
		5: $C \leftarrow iv, c_0, \dots, c_{n-1}$
		6: return C

We would like to show that c is indistinguishable from random strings, just like the one-time pad.

PROOF STRATEGY

1. The One-Time Pad samples $k \leftarrow \{0, 1\}^{n \cdot 128}$, we sample $iv \leftarrow \{0, 1\}^{128}$ and then compute $\text{AES-128}(k, iv + i)$ for $0 \leq i < n$.
 - If we obtain $iv_0 + i = iv_1 + j$ then we got a Two-Time Pad:

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- Even if AES-128 is a PRP, we want a PRF

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Solution This lecture

- Even if AES-128 is a PRP, we want a PRF

Solution Use the PRP-PRF Switching Lemma

PROOF STEP 1: SAMPLING WITHOUT REPLACEMENT

Game ₀	$E(m_0, m_1 \in \{0, 1\}^{128 \cdot n})$	$E_k \left((m_0, \dots, m_{n-1}) \in (\{0, 1\}^{128})^n \right)$
1: $\mathcal{I}, \text{bad} \leftarrow \emptyset, \text{false}$	1: if $ m_0 \neq m_1 $ then	1: $iv \leftarrow \$ \{0, 1\}^{128}$
2: $k \leftarrow \$ \{0, 1\}^{128}; b \leftarrow \$ \{0, 1\}$	2: return $\perp$	2: if $\{iv, \dots, iv + n - 1\} \cap \mathcal{I} \neq \emptyset$ then
3: $b' \leftarrow \mathcal{D}^E$	3: $c \leftarrow \$ E_k(m_b)$	3: $\text{bad} \leftarrow \text{true}$
4: return $b = b'$	4: return c	4: // Game ₁
Game ₁		5: $iv \leftarrow \$$ w.o. overlap with $\mathcal{I}$
1: $\mathcal{I}, \text{bad} \leftarrow \emptyset, \text{false}$		6: $\mathcal{I} \leftarrow \mathcal{I} \cup \{iv, \dots, iv + n - 1\}$
2: $k \leftarrow \$ \{0, 1\}^{128}; b \leftarrow \$ \{0, 1\}$		7: for $0 \leq i < n$ do
3: $b' \leftarrow \mathcal{D}^E$		8: $c_i \leftarrow \text{AES-128}(k, iv + i) \oplus m_i$
4: return $b = b'$		9: return $(iv, c_0, \dots, c_{n-1})$

$$\left| \Pr[\text{Game}_0^{\mathcal{D}}] - \Pr[\text{Game}_1^{\mathcal{D}}] \right| \leq \Pr[\text{Game}_0^{\mathcal{D}} \text{ sets bad}] \leq n \cdot q \cdot (n \cdot q + 1) / 2^{128+1}.$$

PROOF STEP 2: PRP SECURITY OF AES-128

Game₁

```
1:  $\mathcal{I} \leftarrow \emptyset; b \leftarrow \$ \{0, 1\}$   
2:  $k \leftarrow \$ \{0, 1\}^{128}$   
3:  $b' \leftarrow \mathcal{D}^E$   
4: return  $b = b'$ 
```

Game₂

```
1:  $\mathcal{I} \leftarrow \emptyset; b \leftarrow \$ \{0, 1\}$   
2:  $\pi \leftarrow \$$  random permutation  
3:  $b' \leftarrow \mathcal{D}^E$   
4: return  $b = b'$ 
```

$E(m_0, m_1 \in \{0, 1\}^{128 \cdot n})$

```
1: if  $|m_0| \neq |m_1|$  then  
2:   return  $\perp$   
3:  $c \leftarrow \$ E_k(m_b)$   
4: return  $c$ 
```

$E_k \left((m_0, \dots, m_{n-1}) \in (\{0, 1\}^{128})^n \right)$

```
1:  $iv \leftarrow \$$  w.o. overlap with  $\mathcal{I}$   
2:  $\mathcal{I} \leftarrow \mathcal{I} \cup \{iv, \dots, iv + n - 1\}$   
3: for  $0 \leq i < n$  do  
4:    $c_i \leftarrow \text{AES-128}(k, iv + i) \oplus m_i$   
5:    $c_i \leftarrow \pi(iv + i) \oplus m_i // \text{Game}_2$   
6: return  $(iv, c_0, \dots, c_{n-1})$ 
```

Claim: $\left| \Pr[\text{Game}_1^{\mathcal{D}}] - \Pr[\text{Game}_2^{\mathcal{D}}] \right| \leq \text{Adv}_{\text{AES-128}}^{\text{PRP}}(\mathcal{B})$

PROOF STEP 2: PRP SECURITY

Game ₀	P(x)
$\pi \leftarrow \emptyset; k \leftarrow \$ \mathcal{K}$	if $x \notin \pi.\text{keys}$ then
return $\mathcal{D}^p$	$\pi[x] \leftarrow \$ \{0, 1\}^n \setminus \pi.\text{values}$
Game ₁	
	$y \leftarrow \pi[x]$
1: $\pi \leftarrow \emptyset$	$y \leftarrow E_k(x)$ //Game ₀
2: return $\mathcal{A}^p$	return y

$$\text{Adv}_E^{\text{prp}}(\mathcal{A}) := |\Pr[\mathcal{A}^{\text{Game}_0}] - \Pr[\mathcal{A}^{\text{Game}_1}]|$$

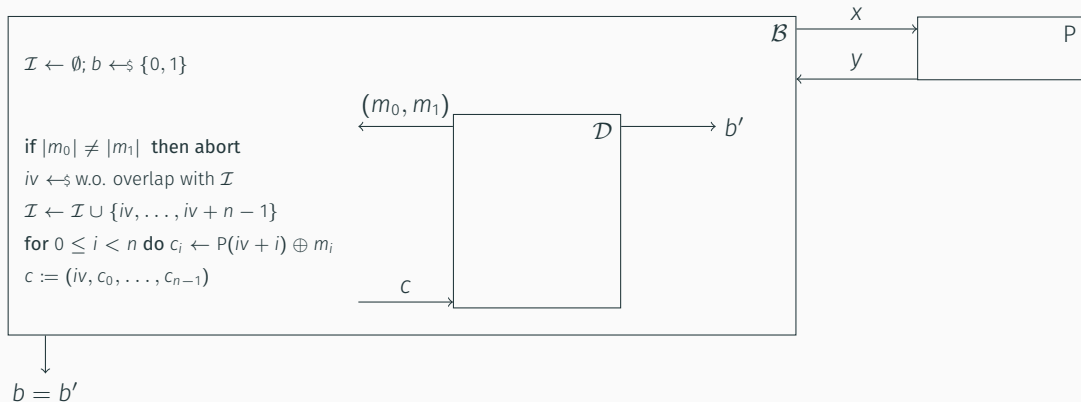
Proof Sketch

- Assume that there is some adversary $\mathcal{D}$ that detects this change and behaves differently in Game₁ and Game₂ of our main proof.
- We use this adversary as a blackbox subroutine in a new adversary $\mathcal{B}$ to break the PRP security of AES-128.

AES-128 Assumption

$\text{Adv}_{\text{AES-128}}^{\text{prp}}(\mathcal{A}) \leq \varepsilon$ for any adversary $\mathcal{A}$ running in time t and $\log(t/\varepsilon) \approx 128$.

PROOF STEP 2: PRP SECURITY (VISUALISATION)



PROOF STEP 2: PRP SECURITY

We use this adversary as a blackbox subroutine in a new adversary $\mathcal{B}$ to break the PRP security of AES-128.

$\mathcal{B}^P$	$E(m_0, m_1 \in \{0, 1\}^{128 \cdot n})$	$E_k \left((m_0, \dots, m_{n-1}) \in (\{0, 1\}^{128})^n \right)$
1: $\mathcal{I} \leftarrow \emptyset; b \leftarrow \{0, 1\}$	1: if $ m_0 \neq m_1 $ then	1: $iv \leftarrow \text{w.o. overlap with } \mathcal{I}$
2: $b' \leftarrow \mathcal{D}^E$	2: return $\perp$	2: $\mathcal{I} \leftarrow \mathcal{I} \cup \{iv, \dots, iv + n - 1\}$
3: return $b = b'$	3: $c \leftarrow E_k(m_b)$	3: for $0 \leq i < n$ do
	4: return c	4: $c_i \leftarrow P(iv + i) \oplus m_i$
		5: return $(iv, c_0, \dots, c_{n-1})$

P is AES-128

$\mathcal{B}^P$ perfectly simulates Game₁

P is random permutation

$\mathcal{B}^P$ perfectly simulates Game₂

PROOF STEP 2: PRP SECURITY

- P is AES-128: $\mathcal{B}^P$ perfectly simulates Game₁
- P is **random permutation**: $\mathcal{B}^P$ perfectly simulates Game₂

Thus if $\mathcal{D}$ significantly differs in its behaviour between games Game₁ and Game₂, $\mathcal{B}$ significantly differs between AES-128 and a random permutation, i.e. it distinguishes

- We obtain: $\left| \Pr[\text{Game}_1^{\mathcal{D}}] - \Pr[\text{Game}_2^{\mathcal{D}}] \right| \leq \text{Adv}_{\text{AES-128}}^{\text{PRP}}(\mathcal{B})$
- We assumed $\text{Adv}_{\text{AES-128}}^{\text{PRP}}(\mathcal{A}) \leq \varepsilon$ for **any** adversary $\mathcal{A}$ running in time t s.t. $\log(t/\varepsilon) \approx 128$.
- Thus: $\left| \Pr[\text{Game}_1^{\mathcal{D}}] - \Pr[\text{Game}_2^{\mathcal{D}}] \right| \leq \varepsilon$.

PROOF STEP 3: PRP-PRF SWITCHING LEMMA

Game ₂	$E(m_0, m_1 \in \{0, 1\}^{128 \cdot n})$	$E_k \left((m_0, \dots, m_{n-1}) \in (\{0, 1\}^{128})^n \right)$
1: $\mathcal{I} \leftarrow \emptyset; b \leftarrow \$ \{0, 1\}$	1: if $ m_0 \neq m_1 $ then	1: $iv \leftarrow \$$ w.o. overlap with $\mathcal{I}$
2: $\pi \leftarrow \$$ random permutation	2: return $\perp$	2: $\mathcal{I} \leftarrow \mathcal{I} \cup \{iv, \dots, iv + n - 1\}$
3: $b' \leftarrow \mathcal{D}^E$	3: $c \leftarrow \$ E_k(m_b)$	3: for $0 \leq i < n$ do
4: return $b = b'$	4: return c	4: $r_i \leftarrow \pi(iv + i)$
Game ₃		5: $r_i \leftarrow \rho(iv + i)$ // Game ₃
1: $\mathcal{I} \leftarrow \emptyset; b \leftarrow \$ \{0, 1\}$		6: $c_i \leftarrow r_i \oplus m_i$
2: $\rho \leftarrow \$$ random function		7: return $(iv, c_0, \dots, c_{n-1})$
3: $b' \leftarrow \mathcal{D}^E$		
4: return $b = b'$		

$$\left| \Pr[\text{Game}_2^{\mathcal{D}}] - \Pr[\text{Game}_3^{\mathcal{D}}] \right| \leq n \cdot q \cdot (n \cdot q + 1) / 2^{128+1}.$$

PROOF: PUTTING IT ALL TOGETHER

$$\begin{aligned}\text{Adv}_{\text{AES-128-CTR}}^{\text{ind-cpa}}(\mathcal{D}) &\leq n \cdot q \cdot (n \cdot q + 1) / 2^{128+1} \\ &\quad + \text{Adv}_{\text{AES-128}}^{\text{prp}}(\mathcal{A}) \\ &\quad + n \cdot q \cdot (n \cdot q + 1) / 2^{128+1} \\ &= \frac{2 n \cdot q \cdot (n \cdot q + 1)}{2^{128+1}} + \text{Adv}_{\text{AES-128}}^{\text{prp}}(\mathcal{A})\end{aligned}$$

sampling w.o. replacement

PRP security

PRP-PRF switching lemma

BOUNDS

- If we allow $n \cdot q = 2^{64}$ then AES-128-CTR offers no security guarantees
- If we allow $n \cdot q = 2^{32}$ then AES-128-CTR offers ≈ 64 “bits of security”

“_("/)_/”

This situation is not unusual!^a

^aAES-256 will not save you here, proof left as homework.

Natural question: is reduction “tight”?

We can **prove** this bound, but is there an attack matching this bound that e.g. breaks IND-CPA with $n \cdot q \approx 2^{64}$ queries?

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We can **prove** this bound, but is there an attack matching this bound that e.g. breaks IND-CPA with $n \cdot q \approx 2^{64}$ queries?

1. Make 2^{64} calls to $E(m_0, m_1)$, call the list of output L_a
2. Make 2^{64} calls to $E(m_0, m_0)$, call the list of output L_b
3. By the “Birthday Paradox” there is a good chance that there exists some $iv_a \in L_a$ and some $iv_b \in L_b$ s.t. $iv_a = iv_b$
4. Check if the matching ciphertexts agree (return $b = 0$) or not (return $b = 1$).

REMINDER: IND-CPA $\neq$ IND-CCA

IND-CCA	$E(m_0, m_1)$	$D(c)$
1: $\mathcal{C} \leftarrow \emptyset$	1: if $ m_0 \neq m_1 $ then	1: if $c \in \mathcal{C}$ then
2: $k \leftarrow \$ \mathcal{K}$	2: return $\perp$	2: return $\perp$
3: $b \leftarrow \$ \{0, 1\}$	3: $c \leftarrow \$ E_k(m_b)$	3: return $D_k(c)$
4: $b' \leftarrow \mathcal{D}^{E,D}$	4: if $m_0 \neq m_1$ then $\mathcal{C} \leftarrow \mathcal{C} \cup \{c\}$	
5: return $b = b'$	5: return c	

$$\text{Adv}_{E,D}^{\text{ind-cca}}(\mathcal{D}) = \left| \Pr[\text{IND-CCA}^{\mathcal{D}} = 1] - 1/2 \right|.$$

REMINDER: IND-CPA $\neq$ IND-CCA

We only proved IND-CPA security, AES-128-CTR (as is) is trivially insecure against an IND-CCA adversary.

Break it!

PRP $\rightarrow$ IND-CPA $\Rightarrow$ SPRP $\rightarrow$ IND-CCA I

Game ₀	P(x)
1: $\pi \leftarrow \emptyset$	1: if $x \in \pi.\text{keys}$ then
2: return $\mathcal{D}^P$	2: $y \leftarrow \pi[x]$
Game ₁	3: else
1: $\pi \leftarrow \emptyset; k \leftarrow \$ \mathcal{K}$	4: $y \leftarrow \$ \{0, 1\}^n \setminus \pi.\text{values}$
2: return $\mathcal{D}^P$	5: $\pi[x] \leftarrow y$
	6: $y \leftarrow E_k(x)$ //Game ₁
	7: return y

Figure 1: PRP Security Games.

PRP $\rightarrow$ IND-CPA $\Rightarrow$ SPRP $\rightarrow$ IND-CCA II

Game ₀	P(x)	P ⁻¹ (y)
1: $\pi \leftarrow \emptyset$	1: if $x \in \pi.\text{keys}$ then	1: if $y \in \pi.\text{values}$ then
2: return $\mathcal{D}^{P,P^{-1}}$	2:	2: // Find the x s.t. $\pi[x] = y$
Game ₁	3: $y \leftarrow \pi[x]$	3: $x \leftarrow \pi^{-1}[y]$
1: $k \leftarrow \$ \mathcal{K}$	4: else	4: else
2: return $\mathcal{D}^{P,P^{-1}}$	5: $y \leftarrow \$ \{0, 1\}^n \setminus \pi.\text{values}$	5: $x \leftarrow \$ \{0, 1\}^n \setminus \pi.\text{keys}$
	6: $\pi[x] \leftarrow y$	6: $\pi[x] \leftarrow y$
	7: $y \leftarrow E_k(x)$ //Game ₁	7: $x \leftarrow E_k^{-1}(y)$ //Game ₁
	8: return y	8: return x

Figure 2: SPRP Security Games.

CRYPTANALYSIS TARGET: $\text{Adv}_{\text{AES-128}}^{(s)\text{prp}}(\mathcal{A})$

Our reduction tells us that if AES-128 is a PRP and if we do not allow too many queries then AES-128-CTR is IND-CPA secure

Reduced scope We can focus our efforts on studying AES-128

Wider scope Any result distinguishing AES-128 from a PRP invalidates our proof

Biclique Cryptanalysis of the Full AES

Andrey Bogdanov*, Dmitry Khovratovich, and Christian Rechberger*

K.U. Leuven, Belgium; Microsoft Research Redmond, USA; ENS Paris and Chaire France Telecom, France

August 31, 2011

Abstract. Since Rijndael was chosen as the Advanced Encryption Standard (AES), improving upon 7-round attacks on the 128-bit key variant (out of 10 rounds) or upon 8-round attacks on the 192/256-bit key variants (out of 12/14 rounds) has been one of the most difficult challenges in the cryptanalysis of block ciphers for more than a decade. In this paper, we present the novel technique of block cipher cryptanalysis with bicliques, which leads to the following results:

- The first key recovery method for the full AES-128 with computational complexity $2^{126.1}$.
- The first key recovery method for the full AES-192 with computational complexity $2^{189.7}$.
- The first key recovery method for the full AES-256 with computational complexity $2^{254.4}$.
- Key recovery methods with lower complexity for the reduced-round versions of AES not considered before, including cryptanalysis of 8-round AES-128 with complexity $2^{124.9}$.
- Preimage search for compression functions based on the full AES versions faster than brute force.

In contrast to most shortcut attacks on AES variants, we *do not need to assume related-keys*. Most of our techniques only need a very small part of the codebook and have low memory requirements, and are practically verified to a large extent. As our cryptanalysis is of high computational complexity, it does not threaten the practical use of AES in any way.

AES-128 IS NOT SPRP-SECURE FOR $\log(t/\epsilon) \geq 126.21$



Andrey Bogdanov, Dmitry Khovratovich, and Christian Rechberger. **Biclique Cryptanalysis of the Full AES**. In: *ASIACRYPT 2011*. Ed. by Dong Hoon Lee and Xiaoyun Wang. Vol. 7073. LNCS. Springer, Berlin, Heidelberg, Dec. 2011, pp. 344–371. DOI: 10.1007/978-3-642-25385-0_19

FIN

READ UP ON IV REUSE ATTACKS!

- [BKR11] Andrey Bogdanov, Dmitry Khovratovich, and Christian Rechberger. **Biclique Cryptanalysis of the Full AES**. In: *ASIACRYPT 2011*. Ed. by Dong Hoon Lee and Xiaoyun Wang. Vol. 7073. LNCS. Springer, Berlin, Heidelberg, Dec. 2011, pp. 344–371. DOI: [10.1007/978-3-642-25385-0_19](https://doi.org/10.1007/978-3-642-25385-0_19).